Section 21.12 Networking

Contents

1. INTRODUCTION

With the GYROSCAN Intera R7.1 a network including an ethernet switch, is part of the basic system delivery. An ethernet switch is in fact a repeater with a hub.
Additional network information is provided to explain the different network configurations that are possible in the Gyroscan Intera environment.

Chapter 2 describes the specific GYROSCAN Intera network.

Chapter 3 describes the situation of a GYROSCAN Intera not connected to the hospital network.

Chapter 4 describes the situation of a GYROSCAN Intera connected to the hospital network system.

Chapter 5 describes information about configuring the network when all the basic hardware has been installed and the system will be connected to the hospital network.

Chapter 6 describes adding and configuring of remote nodes.

Chapter 7 describes configuring of the various DICOM applications.

Chapter 8 describes general network information.

Appendix A is a checklist which has to be filled in when connecting the Gyroscan Intera to the hospital network.

Appendix B is an overview of the default network settings on a Gyroscan Intera not connected to the hospital network.

This section does not describe how to install the hospital network (or 'radiology network').
The hospital's network may already be present or it may be installed by PMS or another contractor as part of the sales agreement. For the installation procedures described in the following paragraphs, we assume that the radiology network has already been installed in the hospital.

The hospital network administrator should provide the information how to connect the Gyroscan Intera to the radiology network and how to setup the network configuration (nodes, IP addresses etc.).

2. GYROSCAN NETWORK INFORMATION

2.1. NETWORK HARDWARE

The Gyroscan Intera is equipped with an ethernet switch.
The ethernet switch is part of the communication link between the host computer and reconstructor computer.
A prerequisite specification of this link is, when connected to a hospital network, that the hospital network load may not influence the network performance between host computer and reconstructor computer
This is prevented by using a ethernet switch.

In order to create a private network between the host computer and reconstructor, the ethernet switch must be programmed. Programming prevents network access from other units than the host computer to the reconstructor.
Network access from other units than the host to the reconstructor can result in poor network performance between host computer and reconstructor and is therefore disabled.
Also the ethernet switch is programmed to create a private printer for the host computer only. This is called a Virtual local area network (LAN).
Programming of the switch has already been done in the MR factory.

The ethernet switch has an autosense function, which will automatically adapt the network speed on it's output port to the hospital network (10Mb/s or 100 Mb/s).

Refer to section computer in the drawings manual, for connector locations of the ethernet switch.
With the supplied UTP cable, only the straight output of the ethernet switch is supported.

NOTE

The ethernet switch used in the R7.1.1 Intera GYROSCAN system only supports UTP in and outlets.

2.2. NETWORK CONFIGURATION

The following units are standard connected in the standalone GYROSCAN Intera system to the ethernet switch:

- Host computer
- Reconstructor
- Print server

Optionally the following units can be connected to the ethernet switch:

- DICOM node/printer
- EasyVision workstation
- Other

When the EasyVision workstation or DICOM unit is shared with other modalities (e.g. CT), the EasyVision workstation or DICOM unit must be connected 'outside' the switch, separately to the hospital network.

NOTE

From release R7.1 onwards, the DWS and GYROVIEW HR are no longer supported

In the Gyroscan Intera basic configuration, 4 TCP/IP addresses and one DECnet address are used and default configured according Table 1.

Unit	TCP/IP address	DECnet address
Host	192.192.192.201	1.201
Reconstructor	192.192.192.210	Not applicable
Print server	192.192.192.211	Not applicable
Ethernet switch	192.192.192.212	Not applicable

Table 1: Default TCP/IP and DECnet addresses

3. GYROSCAN INTERA, NOT CONNECTED TO THE HOSPITAL NETWORK.

This situation is present when the GYROSCAN Intera system is installed and the system will not be connected to the hospital network.

As an option the following units can be connected to the switch:

- EasyVision workstation
- A DICOM node/printer
- Other

An optional point-to-point hardcopy unit can be connected to the IIM unit.

With the options connected as mentioned above, this situation is called a MR workgroup.

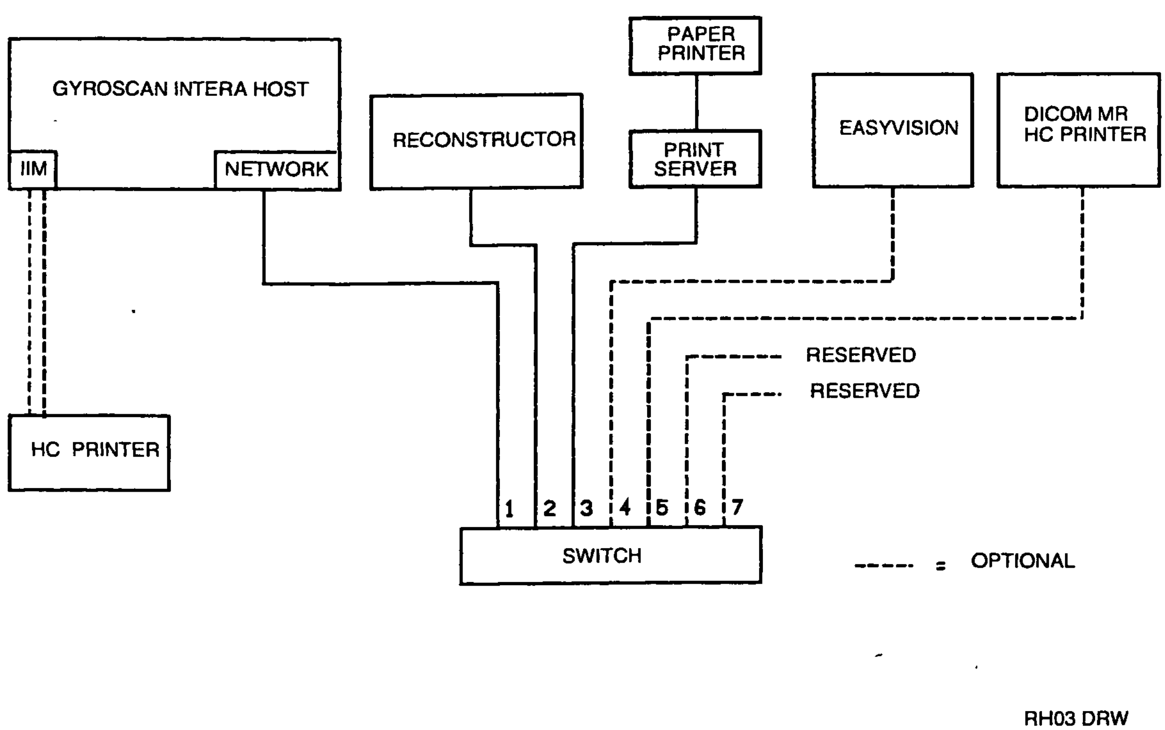

Figure 1: Gyroscan Intera in a MR workgroup

NOTE

The combination of EasyVision release R4 and Gyroscan Intera can support 100 Mb/s ethernet speed. Units, which only support 10MB/s, are also supported by the Gyroscan Intera. Refer to Table 2.

Precondition for a 100 Mb/s ethernet connection within the MR workgroup is the presence of the following computers:

System	Used computer in system
Gyroscan Intera	AlphaServer 800 5/333 or AlphaServer 800 5/500
EasyVision release R4	Ultra-1 or Ultra-2
EasyVision release R4	Sparc-5 with installed 100 BaseT ethernet board (option)
EasyVision release R4	Ultra 5, Ultra 10 or Ultra 60

Table 2: Supported computers for 100BaseT connection

NOTE

When the connected EasyVision workstation or DICOM unit will be shared with other modalities (e.g. CT), the EasyVision workstation or DICOM unit has to be connected outside the switch, separately to the hospital network.
Refer to Figure 2.

Refer to section 6 to add and configure a remote node to the local MR nodes database.
DICOM configuration is described in section 7.

4. GYROSCAN INTERA, CONNECTED TO UTP-BASED HOSPITAL NETWORK

Connection from Gyroscan Intera to a UTP hospital network is done by means of a single UTP cable; this cable is included in the Gyroscan Intera delivery.

CAUTION

At this point, it is not allowed yet to connect the ethernet switch to the hospital network. Connecting the ethernet switch may be done only after completion of the network configuration for hospital connection according chapter 5.

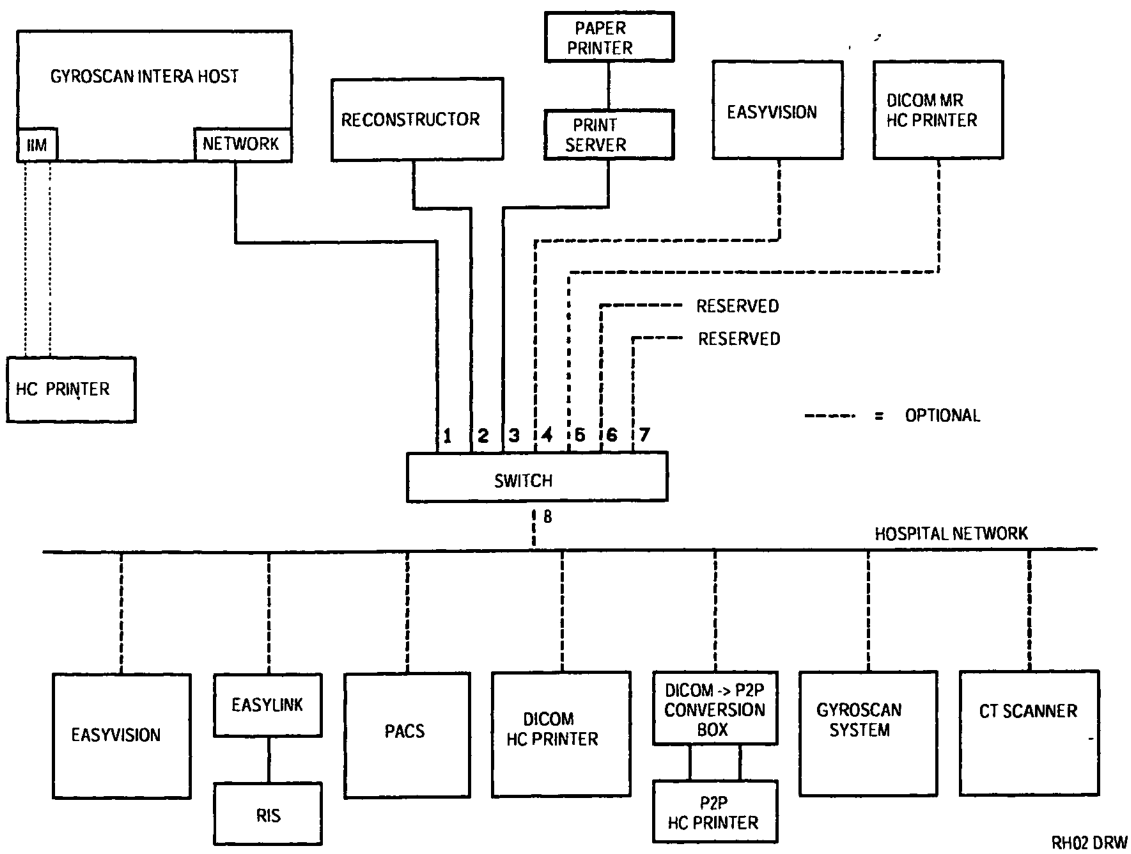

Figure 2: Gyroscan Intera connected to hospital network

Connection from Gyroscan Intera to 10Base5 (Thicknet) hospital network is done by means of a media converter. For a description of this optional installation procedure, read paragraph 4.1.

NOTE

When applicable, a 2^nd^ Gyroscan Intera has to be connected outside the switch, separately to the hospital network.
Connecting the 2^nd^ Gyroscan Intera to a spare ethernet port of the 1^st^ Gyroscan Intera ethernet switch is not supported.

4.1. INSTALLING A MEDIA CONVERTOR

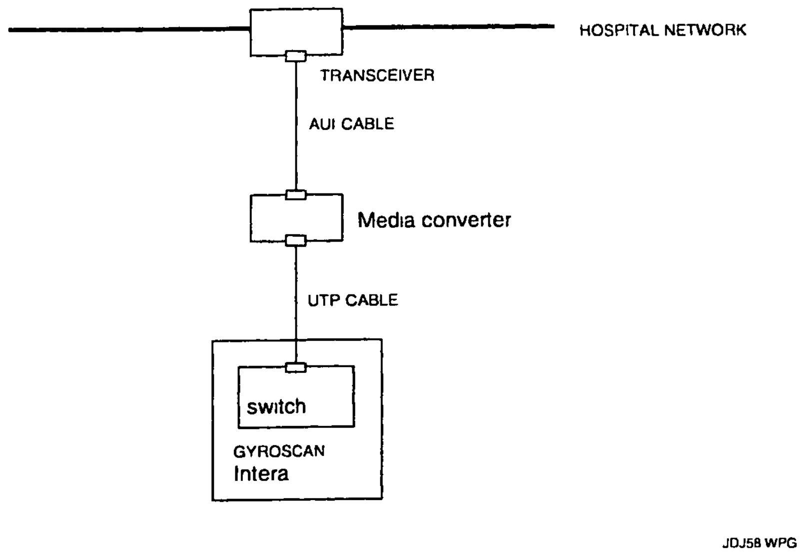

Figure 3: Connecting the media translator

The ethernet switch supports only UTP. If the hospital is equipped with a 10Base5 network, a media converter, interfacing between the hospital network connection and the UTP port on the ethernet switch, should be used.

NOTE

The media converter from UTP to 10Base5 has to be purchased locally. From R7.1 onwards, the PMG MR does not provide these units.

CAUTION

At this point, it is not yet allowed to connect the ethernet switch to the hospital network. Connecting the ethernet switch may be done only after completion of the basic network configuration for hospital connection according chapter 5.

5. CONFIGURING THE NETWORK FOR HOSPITAL CONNECTION

When connected to the hospital network, you should configure the Gyroscan Intera and also the viewing stations or DICOM units that are connected to the Gyroscan Intera. You have to configure the DECnet settings and the TCP/IP settings; both have a different configuration procedure.

NOTE

Contact the local network administrator to obtain unique TCP/IP and DECnet settings before changing the network configuration.
Before changing the local node name and network settings of the host or other units, the checklist according APPENDIX A: Hospital Network LAN Checklist has to be filled in.
Contact the local network administrator to get all the information required.

CAUTION

To complete the network configuration for the basic system successfully, ALL network configuration steps 1-5 have to be executed in the order as mentioned below.

Network configuration steps:

1. Local host
2. Domain name
3. Reconstructor
4. Ethernet switch
5. Print server
6. Connecting the ethernet switch to the hospital network

If you connect an EasyVision workstation to the Gyroscan Intera (within or outside the MR workgroup), also refer to the EasyVision software installation manual.
Refer to chapter 6 to configure the Gyroscan Intera for communication with the EasyVision workstation.
Section 7 describes the configuration of several DICOM applications.

5.1. CONFIGURE LOCAL HOST

The following settings will be used as an example:

New local node name, maximum 6 characters	: MR1
New DECnet node address	: 9.300
New internet address (TCP/IP)	: 120.242.75.120
New Internet Network mask	: 255.255.255.0
New Internet Broadcast mask	: 120.242.75.255
New Internet domain name	: MR

WARNING

Do not change the local network before the checklist is completed!

5.1.1. CONFIGURE THE LOCAL NODE (HOST) DECNET AND TCP/IP SETTINGS:

1. Login on the host with user **system** + <password>.
2. Select: `Network management`
3. Select: `Modify local node name and/or addresses`
4. The system responds with:
   ```
   Tailor configuration of local node
   Enter the node name for this node, <nodename>/?/exit : [NTSCAN]
   ```
5. Enter: **<new nodename>** that you received from the network/system manager. In this example the new node name is MR1.
6. The system responds with:
   ```
   Enter node address (<area>.<number>/EXIT/?> (1.201) :
   ```
7. Enter: **<DECnet node address>**, in this example 9.300, where 9 is the node area and 300 is the nodenumber.
8. The system responds with:
   ```
   Enter internet address for "MR1" :
   ```
9. Enter: **<internet address>** This address has the form of nnn.nnn.nnn.nnn, where 1<= nnn <= 254. In this example 120.242.75.120 has been assigned as the internet number.
10. The system responds with:
    ```
    Enter network mask for "MR1" [255.255.0.0]:
    ```
11. Enter: **<internet network mask>** for this unit. In this example the internet network mask is 255.255.255.0, which will overwrite the given default.
12. The system responds with:
    ```
    Enter broadcast mask for "MR1" [120.242.75.255]:
    ```
13. Enter: **<internet broadcast mask>**, this mask has the form of mmm.mmm.mmm.mmm, where 0 <= mmm <= 255. In this example the given broadcast mask is the same as the one in the checklist and it is sufficient to take the default by pressing **<return>**.
14. When all questions have been answered, the system displays the changes and asks if the databases can be updated.
    The system responds with:

```
Host name                  : MR1
DECnet address             : 9.300
Internet address           : 120.242.75.120
Internet domain            : MR
Internet network mask      : 255.255.255.0
Internet broadcast mask    : 120.242.75.255


Update local node in databases (YES/NO/?) [NO] ?
```

15. Enter: **yes** to update the databases.
16. The system responds with:
    ```
    The system must be rebooted to activate the changes.
    Do you want to reboot now (YES/NO) [YES] ?
    ```
17. Enter: **yes** and the system will shutdown and reboot again twice.

5.2. CONFIGURE DOMAIN NAME

1. Login on the host with user **system** + <password>.
2. Select: `Network management`
3. Select: `Modify local domain`
4. The system responds with:
   ```
   WARNING: the domain name is case sensitive.
   Enclose lowercase names in quotation marks.

   The local node name should be changed first, is required!
   ```

5. `Enter internet domain [MR] :`
6. Enter the new domain name when applicable.
7. Press [RETURN] to go back to the Network Management menu.
8. Select: `Exit to Gyroscan manager Menu`

5.3. CONFIGURE RECONSTRUCTOR

5.3.1. RECONSTRUCTOR CONFIGURATION ON RECONSTRUCTOR

Configuring the host computer TCP/IP address:

1. Switch on the LCD in the drawer of the NTDAC.
2. Login on the reconstructor with user **MRSYSTEM**, password is manager.
3. Select: **Start**
4. Select: **MR configuration**
5. Select: **Host configuration**
6. Enter the TCP/IP address of the host computer: **xxx.xxx.xxx.xxx**

Configuring the reconstructor TCP/IP address:

1. Select: **Start**
2. Select: **MR configuration**
3. Select: **Network**
4. Select: **Identification tab**
5. Click on button : **change**
6. Enter the host name : **RECONSTRUCTOR** (if not already filled in)
7. Click on : **OK**
8. Select: **Protocol tab**
9. Click on: **properties**
10. Enter the TCP/IP address of the reconstructor: **xxx.xxx.xxx.xxx** (in the IP address box)
11. Click on: **OK**
12. Click on: **OK**
13. In some cases the reconstructor has to be rebooted.
    Click on: **Yes** (after the message: `Do you want to restart your computer now?`) when applicable.
    The system will be restarted automatically.
14. Press <Control>+ALT+<Delete>.
15. Select: Lock Workstation.
16. Switch off the LCD in the drawer of the NTDAC.
17. Slide the drawer back in the NTDAC.

5.3.2. RECONSTRUCTOR CONFIGURATION ON HOST

1. Login on host computer with user **system** + <password>.
2. Select: `Configure Hardware Options`
3. Select: `Reconstructor Management menu`
4. Select: `Modify Network Configuration Reconstructor`
5. Enter: **<new nodename>** that you received from the network/system manager. In this example the default nodename RECONSTRUCTOR is selected.
   `Enter the host-name of the Reconstructor [RECONSTRUCTOR]:`
6. Enter the new TCP/IP address and press **<Return>** after:
   `Enter internet address for RECONSTRUCTOR []:` **xxx.xxx.xxx.xxx**
7. Press **<Return>** after:
   `Press [RETURN] for Reconstructor Management menu`
8. Select: `Exit to configure Hardware options menu`

5.4. CONFIGURE ETHERNET SWITCH

CAUTION

Follow ALL the steps (5.4.1 to 5.5) in the order as mentioned below to complete the ethernet switch configuration successfully.

5.4.1. CONFIGURE ETHERNET SWITCH ON HOST

1. Login on the host computer with user **system** + <password>.
2. Select: `Configure Hardware Options`
3. Select: `Install Network switch`
4. Select: `Compaq Netelligent 5708 TX`
5. Enter. `MAC-address Switch:`
   `Enter the MAC-address (Ethernet address) of the 5708 TX Switch`
6. `MAC-address (nn-nn-nn-nn-nn-nn/QUIT/?) [nn-nn-nn-nn-nn-nn]:` **`nn-nn-nn-nn-nn-nn`**

NOTE

The MAC address of the ethernet switch can been found on a label attached at the front, on the right lower corner of the switch.

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7. Enter **<new nodename>** that you received from the network/system manager. In this example the default nodename SWITCH is selected.
   `Enter the host-name of the Switch [SWITCH]:`
8. Enter the new TCP/IP address and press **<Return>** after:
   `Enter internet address for SWITCH []:` **xxx.xxx.xxx.xxx**
9. ```
   The following switch will be installed:
   Switch type: Compaq Netelligent 5708 TX
   MAC-Address Compaq Netelligent 5708 TX: nn-nn-nn-nn-nn-nn
   Nodename Compaq Netelligent 5708 TX: SWITCH
   TCP/IP address Compaq Netelligent 5708TX: xxx.xxx.xxx.xxx
   ```
10. Press **<Return>** to confirm the settings:
    ```
    Is this information correct? (YES/NO) [YES]:
    Application SW is stopped...
    Stopping server GYROSCAN

    TCP/IP network is restarted...
    ```
11. Press **<Return>** after:
    `Press [Return] for main menu.`

5.4.2. RESET ETHERNET SWITCH TO DEFAULTS

In order to guarantee a correct IP address in the ethernet switch, the ethernet switch settings have to be reset to defaults. The reset is followed by an automatic reboot of the switch. After the reboot, the ethernet switch will load the correct IP address from the host.

1. Login on the host computer with user **system** + <password>.
2. Select: `Configure Hardware Options`
3. Select: `Install Network switch`
    ```

4. Select: `RS232 connection to Network switch`
5. The switch will respond with the following login screen:

```
%DCL-I-ALLOC, _NTSCAN$TTB0: allocated

%REM-I-TOQUIT, connection established

Press Ctrl/\ to quit, Ctrl/@ for command mode
```

6. Enter the password of the switch to open the main menu: **admin <Return>**
7. The switch will respond with the main menu.
8. Select **Reset...** menu with the cursor keys and **press <Return>.**
9. The Reset menu will be displayed.
10. Select item: **Reset EEprom to Factory Default Value** with the cursor keys and **press <Return>**
11. The switch will respond with:
    `Are you sure? No`
12. Select **Yes** with the **right arrow key** and press <Return>
13. The switch will respond with:

```
System is writing factory default values to EEPROM.
............
```

14. After a few minutes, the switch will boot again while displaying the powerup selftest results:

```
Netelligent 5708 Ethernet Switch Version 1.91
-  Initialization started
-  Boot initialization started.
RS232 test - - - - - - - - - - - - -   OK!
NIC Test
    NIC ID test - - - - - - - - - - -  OK!
    NIC Loopback test - - - - - - - -  OK!
    NIC Interrupt test- - - - - - - -  OK!
EEPROM test  - - - - - - - - - - - -   OK!
FLASH test - - - - - - - - - - - - -   OK!
AAT&C1&D2S0=1V1
AAT&C1&D2S0=1V1
AAT&C1&D2S0=1V1
```

5.4.3. CONFIGURE VIRTUAL LAN ON ETHERNET SWITCH

1. At the end of the powerup-sequence of the ethernet switch, the login screen is displayed again via the RS232 link.

```
                 Compaq Computer Corporation


                 Netelligent 5708 TX

(c) Copyright 1997 Compaq Computer Corporation. All rights reserved.
Hardware Main Board Version 2
Firmware Main Board Version 2.38
Hardware Agent Module Version 1
Firmware Agent Module Version 2.15
Password:
```

2 Enter the password of the switch to open the main menu: **admin <Return>**
3. The switch will respond with the main menu:
4. Select `Virtual LAN Configuration` from the Main Menu.
5. This will show the following menu.
6. Select `Port Grouping` with the arrow up key and **press <Return>**
7. As a result, the (default) Port grouping Configuration will be displayed.

```
                 Port Grouping Configuration

Port Group1 Group2 Group3 Group4 Group5 Group6 Group7 Group8
------------------------------------------------------------
1     Yes    No     No     No     No     No     No     No
2     Yes    No     No     No     No     No     No     No
3     Yes    No     No     No     No     No     No     No
4     Yes    No     No     No     No     No     No     No
5     Yes    No     No     No     No     No     No     No
6     Yes    No     No     No     No     No     No     No
7     Yes    No     No     No     No     No     No     No
8     Yes    No     No     No     No     No     No     No

      Return to Previous Menu

Use cursor keys to choose item. Press <ENTER> to confirm choice.
Press <CTRL><N> to return to the Main Menu
```

8. The above configuration is the result of setting the switch to factory defaults. Follow the next steps to configure it as required for the MR-system.
9. **Notes**:
   - Start with the configuration of the ports for Group2 and then the ports for Group1. This will avoid warning messages like: `"Each port should at least belong to one group"`
   - You can ignore the message: `"STA would not run because of multiple groups in the switch."`
10. To assign a port to a particular port group, highlight the group number for the desired port, by selecting it with the cursor keys and press <Return>.
11. The switch will respond with the following question:

    Do you want to join this group? No
12. Press the left or right arrow key to change the port group entry to **Yes** or **No.**
13. Press <Return> to confirm your choice.
14. Repeat steps 10 to 13 to get the Virtual LAN configuration displayed below.

Port Grouping Configuration

Port	Group1	Group2	Group3	Group4	Group5	Group6	Group7	Group8
1	Yes	**Yes**	No	No	No	No	No	No
2	Yes	No	No	No	No	No	No	No
3	Yes	No	No	No	No	No	No	No
4	Yes	No	No	No	No	No	No	No
5	Yes	No	No	No	No	No	No	No
6	Yes	No	No	No	No	No	No	No
7	Yes	No	No	No	No	No	No	No
8	**No**	**Yes**	No	No	No	No	No	No

15. At the end of the Virtual LAN Configuration press <**CTRL-N**> to get to the main menu.
16. Select LOGOFF . The switch responds with:

    Are you sure? No
17. Select **Yes** with the right arrow key and press <Return>
18. Press **Ctrl ** key's simultaneously to return to Configure Switch menu
19. Select: `Exit to Configure Hardware options Menu`

5.4.4. CONFIGURE ETHERNET SWITCH FOR FULL-DUPLEX OPERATION HOST

Because the switch has been reset to defaults according 5.4.2, port 1 of the switch has to be configured again as a Full duplex port. This modification is nessesary because the network setting of the host computer is also set to 100Mb Full-Duplex.

1. Login on the host computer with user **system** + <password>.
2. Select: `Configure Hardware Options`
3. Select: `Install Network switch`
4. Select: `RS232 connection to Network switch`
   The switch will respond with the following login screen:

```
%DCL-I-ALLOC, _NTSCAN$TTB0: allocated

%REM-I-TOQUIT, connection established

Press Ctrl/\ to quit, Ctrl/@ for command mode
```

5. Enter the password of the switch to open the main menu: **admin <Return>**
6. The switch will respond with the main menu.
7. Select the **`Port configuration..`** menu with the cursor keys and **press <Return>.**
8. The Port configuration menu will be displayed:

Port Configuration

Port	Enabled	Duplex	Aging Time	Switching mode	Flow control
1	Yes	**Full**	300	CT	Yes
2	Yes	Auto	300	CT	Yes
3	Yes	Auto	300	CT	Yes
4	Yes	Auto	300	CT	Yes
5	Yes	Auto	300	CT	Yes
6	Yes	Auto	300	CT	Yes
7	Yes	Auto	300	CT	Yes
8	Yes	Auto	300	CT	Yes

9. Select the Duplex entry for port 1 with the arrow keys.
10. Press <Return> to confirm choice.
11. Press the left or right arrow key to change the Duplex setting for port 1 to **Full.**

12 Press <Return> to confirm choice.
13 After completion of the Port Configuration press **<CTRL-N>** to get to the main menu.
14. Select LOGOFF . The switch responds with:
Are you sure? No
15. Select **Yes** with the right arrow key and press <Return>
16. Press **Ctrl ** key's simultaneously to return to Configure Switch menu
17. Select: `Exit to Configure Hardware options Menu`

5.5. CONFIGURE ETHERNET SWITCH FOR FULL OR HALF-DUPLEX HOSPITAL NETWORK CONNECTION

Table 3 shows the preferred MR system configuration to the ethernet switch of the Gyroscan Intera with respect to the highest performance of the hospital network connection.

Preferred order	Hospital network type	Switch setting port 8
1	100 Mb/s, Full Duplex	**Full**
2	100 Mb/s, Half Duplex	**Half**
3	10 Mb/s, Full Duplex	**Full**
4	10 Mb/s, Half Duplex	**Half**

Table 3: Preferred hospital network setting

Contact the local network administrator in order to configure the highest preferred setting on the hospital network port.
Generally, a Collision Domain (Repeater/Hub) hospital network connection is a network of a Half-duplex type.

1. Login on the host computer with user **system** + <password>.
2. Select: `Configure Hardware Options`
3. Select: `Install Network switch`
4. Select: `RS232 connection to Network switch`
The switch will respond with the following login screen:

```
%DCL-I-ALLOC, _NTSCAN$TTB0: allocated

%REM-I-TOQUIT, connection established

Press Ctrl/\ to quit, Ctrl/@ for command mode
```

5. Enter the password of the switch to open the main menu: **admin <Return>**
6. The switch will respond with the main menu.
7. Select the **`Port configuration`**... menu with the cursor keys and **press <Return>.**
8. The Port configuration menu will be displayed:

```
                   Port Configuration

Port Enabled Duplex  Aging Time  Switching mode  Flow control
--------------------------------------------------------------
1      Yes    Full       300           CT            Yes
2      Yes    Auto       300           CT            Yes
3      Yes    Auto       300           CT            Yes
4      Yes    Auto       300           CT            Yes
5      Yes    Auto       300           CT            Yes
6      Yes    Auto       300           CT            Yes
7      Yes    Auto       300           CT            Yes
8      Yes    Full/Half  300           CT            Yes
```

9. Select the Duplex entry for port 8 with the arrow keys.
10. Press <Return> to confirm choice.
11. Press the left or right arrow key to change the Duplex setting for port 8 according Table 3.
12. Press <Return> to confirm choice.
13. After completion of the Port Configuration press <**CTRL-N**> to get to the main menu.
14. Select LOGOFF The switch responds with:
    Are you sure? No
15. Select **Yes** with the right arrow key and press <Return>
16. Press **Ctrl ** key's simultaneously to return to Configure Switch menu
17. Select: `Exit to Configure Hardware options Menu`

5.6. CONFIGURE PRINT SERVER

1. Switch off the printer.
2. Login on the host computer with user **system** + <password>.
3. Select: `Configure Hardware Options`
4. Select: `Install network printer (+ queue's)`
   The system will display the '*configure printers*' menu.
5. Select: `Supported HP deskjet printer`
6. Select: **<paper size>** after:
   `Enter paper size (A4/LETTER) [A4]`
   `(A4 is the European A4 format, LETTER is the US Quattro format).`
7. Enter: <**MAC address**> (e.g. 00-60-B0-96-CC-87)

NOTE

The MAC address is written on a label, which is attached to the bottom of the printer server and is indicated with the text 'AD'. Refer to Figure 4.

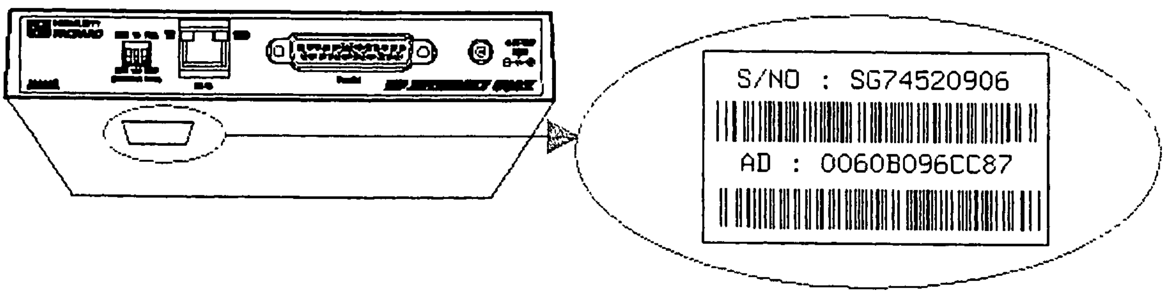

Figure 4: MAC address location print server

7. Enter: **<new nodename>** that you received from the network/system manager. In this example the default node name JETDIRECT is selected.
- Enter the host-name of the JetDirect [JETDIRECT]:
8. Now enter the host name of the printer server
   Enter the internet address for *host-name JetDirect*

9. Enter the new TCP/IP address of the printer server: **xxx.xxx.xxx.xxx**
10. Enter: **Yes** after:
    Is this information correct? (YES/NO) [YES]
11. The print server is now installed and the TCP/IP network is restarted.
12. Press <**Return**> after.
    `Press [RETURN] for main menu`
13.

NOTE

Switch the power of the print server off and on.

---

14. **Wait 2 minutes for the print server to reboot.**
15. Switch on the printer.
16. Printing a test-page can test the printer connection.
    If you want to test the printer, select option **3** *Print test-plot.* There are 2 test-prints available: a graphical test-plot (PCL) and an ASCII test plot.
    The ASCII test-plot suffices for the printer connection test
17. Select: `Exit to Hardware option menu`
18. Select: `Exit to Gyroscan manager Menu`
19. Select: `Logout` and confirm by clicking on **Yes.**

5.7. CONNECTING THE ETHERNET SWITCH TO THE HOSPITAL NETWORK.

At this point, all basic units have been configured with the correct network settings.
Connect the straight output of the ethernet switch to the hospital network with the delivered UTP cable.

6. CONFIGURE ADDITIONAL REMOTE NODES

Before you add a node to the local MRNODES database the following checklist must be filled out

Node name, maximum 6 characters	
DECnet node address	
Username of IMPORT/EXPORT account	
Datalink task taskname on remote node	
Datalink function for 'local' node	
IMPORT/EXPORT account on remote node	
Allow remote access by 'local' node users	

As an example, paragraph 6.1 and 6.2 describes the configuration of an EasyVision workstation.

Repeat all the steps according paragraph 6.1 to configure other additional nodes to the MRNODES database.

6.1. CONFIGURE THE EASYVISION

NOTE

The taskname "GCOMSERVER" implies storing of the images into the database of the EasyVision. Using another taskname instead of "GCOMSERVER" makes it possible to specify other functions on the EasyVision according Table 4. ***Repeat the steps 1-17 when applicable.***

The following configuration example will be used to describe the configuration steps:

Node name, maximum 6 characters	: EV1	
DECnet node address	: 9.310	
Enter username of IMPORT/EXPORT account	: default	[ ]
Datalink task taskname on remote node	: default	[GCOMSERVER]
Datalink function for 'local' node	: default	YES
IMPORT/EXPORT account on remote node	: default	YES
Allow remote access by 'local' node users	: default	YES

WARNING

DO NOT CHANGE THE LOCAL NETWORK BEFORE THE CHECKLIST IS COMPLETED !!

After filling in the checklist, execute the following steps:

1. Login on host computer with user **system** + <password>.
2. Select: `Network management`
3. Select: `Add remote node to local MRNODES database`
4. The system responds with:

Add node to local MRNODES and /etc/hosts databases
Enter node name (<name>/EXIT/?) [EXIT] :

5. Enter: **<new nodename>**, that you received from the network administrator (see checklist and refer if applicable to APPENDIX A: Hospital Network LAN Checklist). In this example the new node name is EV1 (A maximum of 6 characters is allowed!)
6. The system responds with:
   Enter node address (<area>.<number>/EXIT/?>) [aa.nnn] :
7. Enter: **<DECnet node address>**. In this example 9.310, where 9 is the node area and 310 is the node number.
8. The system responds with:
   Enter username of IMPORT/EXPORT account on remote node (<name>/?) []:
   In this example the default can be taken.
9. The system responds with:
   Enter DATALINK taskname on remote node (<name>/?) [GCOMSERVER]:
   Here too, for this example, the default can be taken.
10. The system responds with:
    Enable DATALINK function for "EV1" (YES/NO/?) [YES] ?
    Here too, for this example, the default can be taken.
11. The system responds with:
    Enable IMPORT/EXPORT functions for "EV1" (YES/NO/?) [YES] ?
    Here too, for this example, the default can be taken
12. The system responds with:
    Allow remote access by "EV1" users (YES/NO/?) [YES] ?
    Here too, for this example, the default can be taken.
13. Now the system displays the values entered and asks if the node definitions can be stored:

```
Nodename  address   Gview  Username  Password  Taskname    DL   IE   Access
EV1    0  9.310     NO                         GCOMSERVER  YES  YES  YES

Store above node definition (YES/NO/? [] ?
```

14. Enter: **Yes**.
    If the response is YES, the system stores all values into the MRNODES database.
15. The system responds with:
    Press [RETURN] for main menu
16. Press **<Return>**
17. Select: Exit to Gyroscan manager menu

Tasknames	Function
GCOMSERVER	The images are stored into the database of the EasyVision.
GCOM_P	Print: the EasyVision also automatically prints the images.
GCOM_F	File: the images are also automatically written to optical disk.
GCOM_E	Export: the images are also automatically transferred to an external node as customized on the EasyVision.

Table 4: GCOM taskname overview

The combined extensions: GCOM_PF, _PE, _FE or _PFE, activate the corresponding functions simultaneously.

6.2. CONFIGURE PIPELINE QUOTA FOR EASYVISION

To prevent image transfer performance problems to the EasyVision, it is necessary to limit the pipeline quota on the host computer.

1. Login on the host with user **system** + <password>.
2. Select: `Exit to DCL`
3. Enter: **ncp**
4. Enter: NCP> **set executor pipeline quota 0**
5. Enter: NCP> **define executor pipeline quota 0**
6. Enter: **Exit**
7. Enter: **Logout** and confirm by clicking 'Yes"

6.3. SELECT PUSH-NODE

When a workstation (e.g. EasyVision) has been added to the MRNODES database, it is possible to define this workstation as a push-node.

Procedure:

1. Login on the host computer with user **Gyroscan.**
2. Select: **Scan control**
3. Select: **Scan Utility**
4. Select: **Select Push-node**
5. Select applicable push-node.
6. Select: **<Proceed>**
7. Select: System
8. Select: Exit
9. Select. **<Proceed>** to confirm.

7. CONFIGURE DICOM

The Digital Imaging and Communications in Medicine (DICOM) Standard is a detailed specification that describes the means of formatting and exchanging images and associated information.
The standard applies to the operation of the interface which is used to transfer data from and to an imaging device
DICOM is the result of an alliance of users of the standard (members of the American College of Radiology-ACR) with the companies that manufacture medical equipment (members of the National Electrical Medical Association – NEMA).
DICOM relies on computer industry standard network connections and addresses more effectively the communication of digital images from diagnostic modalities such as CT, MR, nuclear medicine and ultrasound (contributions from corresponding NEMA sections).
Besides this also computerized radiography, digitized film, video capture and HIS/RIS information is addressed
It also supports the connection of network hardcopy devices.

The GYROSCAN Intera system supports the following DICOM functionality:

DICOM EXPORT (Store) user

- a DICOM Image Export function to transfer DICOM MR images and the image related data from the GYROSCAN to a remote system.

DICOM IMPORT provider

- a DICOM Image import function to transfer DICOM MR images and the image related data from a remote system to the GYROSCAN system.

DICOM RIS (with Easylink) (software license key needed)

- a DICOM Radiology Information System (RIS) interface to retrieve the patient information for the next examination from the worklist to be examined.

DICOM PRINT user

- a DICOM print function to transfer DICOM MR images and the image related data to a DICOM HC printer.

7.1. DICOM AETITLE CONVENTIONS

NOTES:

The DICOM AETitles and IP hostnames are CASE SENSITIVE !

The DICOM AETitle must be unique per site.
This means, in case of more systems, you have to fill in different names.

For the AETitle the characters a-z, A-Z, 0-9 , and underscore are allowed, spaces and special characters not.

7.2. CONFIGURE GYROSCAN DICOM IMPORT/EXPORT

To be able to use the DICOM Import/export functionality, the following is required:

- Configure the GYROSCAN DICOM Import and/or Export AETitle.
- Define the DICOM Import/Export node.
- Restart the processes using GYRORESTART

7.2.1. CONFIGURE THE DICOM AETITLE

1. Login on the host computer with user **GYROTOOL**
2. Select: Modify System Configuration
3. Select: DICOM parameters
4. Select: DICOM AETITLES
5. Adapt the STORE export AETitle.
   The default Gyroscan DICOM STORE EXP AETitle is. **MR_STORE_EXP**
6. Adapt the STORE import AETitle.
   The default Gyroscan DICOM STORE IMP AETitle is: **MR_STORE_IMP**
7. Press: **<Proceed>**
8. Press: **<Exit>** to leave Figman
9. Select: Exit from GYROTOOL

7.2.2. DEFINE A DICOM IMPORT/EXPORT/RIS NODE

In this example, a DICOM export node is defined.

1. Login on host computer with user **system** + <password>.
2. Select: `DICOM Management`
3. Select one of the applicable entries:
   - `DICOM Export Service Class User`
   - `DICOM Import Service Class Provider`
   - `DICOM RIS Service Class User`
4. Select· `DICOM Export Service Class User`
5. Select: `Add DICOM SCP/SCU`

CAUTION

The AE title and hostname you use in this part, must be the same as the AE title entered at the remote node.
These names are case sensitive.

6. The system responds with e.g.:

```
Add DICOM Export Service Class User
DICOM application                    Hostname      Portnumber
----------------------------------------------------------------
dicom_storage                        APOLLO           5300
```

7. `Enter DICOM application entity to add(<name>/EXIT/?)   [EXIT]:`
8. Enter: **EV1** (or the DICOM application entity title of the remote node).

9.

NOTE

Only a-z, A-Z, 0-9 and - are allowed, spaces, underscores and other characters are not allowed.

---

10. The system responds with:
    `Enter hostname (<name>/EXIT/?) [EXIT]:`
11. Enter: **EasyVision** (or the remote DICOM hostname), if it is a new application in your network Otherwise you should enter the **TCP/IP host name** of the (existing) remote host that supports your DICOM transfer.
12. The system responds with:
    `Enter portnumber (<number>/EXIT/?) [EXIT]:`
13. Enter: **3010** (the portnumber where EasyVision listens to).
    In this example the default portnumber for DICOM export for the EasyVision is used.
14. Now the system displays the values entered and asks if the node definitions can be stored, e g.:

```
DICOM application entity      : EV1         (name of the application)
Host                          : EASYVISION  (the characters of this name will be changed
                                                 automatically in capitals)
Portnumber                    : 3010

Is this correct?(YES/NO) [YES]:
```

15. Enter: **Yes.**
    Now the system stores all values.
16. If the hostname does not exist in the TCP/IP host database, the system will ask for the TCP/IP address.
    `ENTER IP address for host EASYVISION (<IP number>/EXIT) [EXIT]:`
17. Enter: **<IP address>**
18. After this the system responds with:
    `Press [RETURN] for DICOM MANAGEMENT menu`
19. Press **<Return>** to return to DICOM SCP/SCU menu
20. Select: `Exit to Gyroscan Manager menu`
21. **Repeat steps 2 to 20 if a DICOM import or DICOM RIS node has to be defined.**
22. Select: `Exit to DCL`

NOTE

*Some processes should be restarted, so a GYRORESTART **must** be executed before you can use the configured DICOM nodes.*

7.3. CONFIGURE GYROSCAN RIS DICOM

To be able to use DICOM RIS the following is required:

- RIS Interface (MMR 5641) software license key.
- Define the GYROSCAN as RIS DICOM node (SCU)
- Configure the RIS DICOM node (SCP) via which the patient information is received.
- Select the RIS system.
- Modifying RIS worklist time-out
- Restart the processes using GYRORESTART

The default GYROSCAN RIS DICOM parameters are.

Application Entity title: **MR_WLM_RIS**
Work list timeout: **20**
Remote AE title: **MR1** (This is the RIS DICOM SCP Application entity)

These are the parameters that should be configured on the RIS DICOM SCP.
In order to communicate, both the MR scanner and the RIS system should know how to find each other on the network.
This is done via TCP/IP addresses, Application entity (AE) titles and host names.
Before you start configuring, complete the table listed below.
If your system is part of a large RIS network, contact the responsible network administrator.

	Gyroscan RIS DICOM (SCU)	**RIS DICOM (SCP)**
Host Name	Gyroscan TCP/IP host name	
AE title		
TCP/IP address	Gyroscan TCP/IP address	
Port number	Not applicable	

7.3.1. Define a DICOM RIS node

Refer to paragraph 7.2.2 for this procedure:

1. In the DICOM Management menu, select:
   `DICOM RIS Service Class User`
2. Select: `Add DICOM SCP/SCU`

7.3.2. Configure the DICOM RIS AETitle

1. Login on the host computer with user **GYROTOOL**
2. Select: Modify System Configuration
3. Select: DICOM parameters
4. Select: DICOM AETITLES
5. Adapt the RIS AETitle.
   The default Gyroscan DICOM RIS AETitle is: **MR_WLM_RIS**
6.

NOTE

The DICOM AE title must be unique per site.
This means, in case of more systems, you have to fill in different names

7. Press: **<Proceed>**
8. Press: **<Exit>** to leave Figman
9. Select: **1** - Exit from GYROTOOL

7.3.3. Selecting a RIS system

1. Login on the host computer with user **GYROTOOL**
2. Select: Modify System Configuration
3. Select: DICOM parameters
4. Select: DICOM RIS selection
5. Select the RIS system
6. Press: **<Proceed>**
7. Press: **<Exit>** to leave Figman
8. Select: Exit from GYROTOOL

7.3.4. Modifying RIS worklist time-out

The Worklist Timeout is defined as the maximum time in seconds that the MR DICOM application waits for the worklist response from a DICOM RIS.
This timeout is applicable for applications that retrieve the next examination for the MR modality from a RIS system. The value spans all the communication between MR modality and RIS system.
This time-out parameter can be increased up to 30 seconds.

3. Login on the host computer with user **GYROTOOL**
4. Select: Modify System Configuration
5. Select: DICOM parameters
6. Select: DICOM RIS parameters
7. Adapt the Worklist Timeout parameter:

Parameter	Value	Range
Worklist time out	20	[5-30]

6. Press: <**Proceed>**
7. Press: **<Exit>** to leave Figman
8. Select: Exit from GYROTOOL

NOTE

*Some processes should be restarted, so a GYRORESTART **must** be executed before the configuration changes are effective.*

7.4. Configure RIS DICOM with EasyLink station

To be able to configure the DICOM RIS EasyLink, the following is needed:

- RIS Interface (MMR 5641) software license key.
- Define the GYROSCAN as RIS DICOM node (SCU)
- Configure the EasyLink RIS DICOM node (SCP) to receive patient information from.
- Select the EasyLink RIS DICOM station.
- Restart the processes using GYRORESTART

As an example you will find a step by step procedure that describes how to connect an EasyLink station to a Gyroscan MR system.

	Gyroscan RIS DICOM (SCU)	RIS DICOM (SCP)
Host Name	Gyroscan TCP/IP host name	EL1
AE title	GYRO_RIS	VERYEASY
TCP/IP address	Gyroscan TCP/IP address	192.192.192.203
Port number	Not applicable	3320

7.4.1. DEFINE THE EASYLINK DICOM RIS NODE

1. Login on the host computer with user **system** + <password>.
2. Select: `DICOM Management`
3. Select: `DICOM RIS Service Class User`
4. Select. `Add DICOM SCP/SCU`
5. The system responds: `Enter DICOM application entity (<name/EXIT/?) [EXIT]:`
6. Enter: **VERYEASY**
7. The system responds: `Enter hostname (<name/EXIT/?) [EXIT]:`
8. Enter: **EL1**
9. The system responds: `Enter portnumber (<number/EXIT/?) [EXIT]:`
10. Enter: **3320** (This is the portnumber where EasyVision listens to).
11. Now the system displays the values entered and asks if the node definitions can be stored:

RIS DICOM AE : VERYEASY
Host : EL1
Portnumber : 3320

Is this correct? (YES/NO) [YES] :

12. Enter: **Yes** (The system stores the values)
13. If the hostname does not exist in the TCP/IP host database, the system will ask for the internet address.
`ENTER IP address for host EASYVISION (<IP number/EXIT) [EXIT]:`
14. Enter: **<IP address>**
15. The system responds: `Press [RETURN] for DICOM MANAGEMENT menu`
16. Press **Return** to return to DICOM Service Class User menu
17. Select: `Exit to DICOM Management Menu`
18. Select: `Exit to Gyroscan Manager menu`

7.4.2. CONFIGURE RIS DICOM AETITLE

After the EasyLink station has been defined, the Gyroscan system can identify the RIS DICOM system on the network.
In our example the AE title of the Gyroscan RIS DICOM is different from the default value.

1. Login on the host computer with user **GYROTOOL**
2. Select: Modify System Configuration
3. Select: DICOM parameters
4. Select: DICOM AETITLES

```
-----------------------------------------------------------------------------
Parameter        String value

RIS AETitle            MR_WLM_RIS_______
STORE EXP AETitle  MR_STORE_EXP____
STORE IMP AETitle   MR_STORE_IMP____
|----------------------------------------------------------------------------|
```

NOTE

The GYROSCAN RIS DICOM default AETitle is MR_WLM_RIS
This name must be unique per system, which means that in case of more systems, different names should be configured.

5. Configure the RIS AETitle: **GYRO_RIS**
6. Press: **<Proceed>**
7. Press: **<Exit>** to leave Figman
8. Select· Exit from GYROTOOL

7.4.3. Selecting an Easylink RIS system

9. Login on the host computer with user **GYROTOOL**
10. Select: Modify System Configuration
11. Select: DICOM parameters
12. Select: DICOM RIS selection
13. Select the RIS system
14. Press: **<Proceed>**
15. Press: **<Exit>** to leave Figman
16. Select: Exit from GYROTOOL

NOTE

*Some processes should be restarted, so a GYRORESTART **must** be executed before the configuration changes are effective.*

7.5. Configure DICOM HC printer

7.5.1. Select DICOM HC printer

1. Login on the host with user **GYROTOOL**
2. Select· Modify System Configuration
3. Select: Console configuration
4. Select: Copy facilities
5.

NOTE

If a DICOM selection is made, leave the Point-to-Point camera setting to None, otherwise it will not accept the selection

6. Select "DICOM camera" according to the installed DICOM HC printer.
   In this example the AGFA AG_MG3000 DICOM printer is selected.
7. Select: <**Proceed**>
8. Select: <**Cancel**>
9. Select: Exit

7.5.2. DEFINE DICOM HC PRINTER

In this example the AGFA AG_MG3000 DICOM HC printer is selected.

1. Login on the host computer with user **system** + <password>
2. Select: `DICOM management`
3. Select: `Configure DICOM Printer`
4. The system responds with:
   `Please enter ISGDICOMagfa.scu_ae_title: []:`
   Press <**Return**> in case a single provider DICOM printer is used on the network.

NOTE

If the DICOM printer supports provider functionality for more than one users (MRI 1, MRI 2, etc), a scu_ae_tiltle must be given on each MR system e.g. MR1

5. The system responds with:
   `Please enter ISGDICOMagfa.scp_ae_title [MCL]:`
   Press <**Return**> or a new Service Class Provider Application Entity title, according the DICOM printer setup. In this example the default is selected.
6. The system responds with:
   `Please enter ISGDICOMagfa.port [104]:`
   Enter the portnumber to where the DICOM printer is sensing. In this example the default is selected.
7. The system responds with:
   `Enter internet address for dicom_agfa []:`
8. Enter IP address of the DICOM printer: **xxx.xxx.xxx.xxx**
9. Now the system displays the values entered.

```
LOCAL database

Host address            Host name

xxx.xxx.xxx.xxx         DICOM_AGFA, dicom_agfa
```

10. Press <**Return**> for `DICOM Management Menu`
11. Select: `Exit to Gyroscan Manager menu`
12. Select: **Logout** and confirm by clicking **yes**.

NOTE

*Some processes should be restarted, so a GYRORESTART **must** be executed before the configuration changes are effective.*

7.6. DELETE A DICOM NODE

1. Login on host computer with user **system** + <password>
2. Select: `DICOM Management`
3. Select one of the applicable entries.
   - `DICOM Export Service Class User`
   - `DICOM Import Service Class Provider`
   - `DICOM RIS Service Class User`
4. Select: `Delete DICOM SCP/SCU`
5. The system responds e.g with:

```
Delete
DICOM application             Hostname      Portnumber
--------------------------------------------------------
EV1                           EASYVISION       3010
```

6. `Enter DICOM application entity (<name>/EXIT/?)  [EXIT]:`
7. Enter: **<dicom application name>** (name of application that you want to delete, e.g. EV1)

NOTE

The dicom application name is case sensitive

8. The system responds with:
   `DICOM application entity to delete : EV1`
   `Do you want to continue? (YES/NO) [NO]:`
9. Enter: **yes**.
10. The system responds with:
    `Press [RETURN] for DICOM MANAGEMENT menu`
11. Select: `Exit to DICOM management menu`
12. Repeat steps 2 to 11 when applicable.
12. Select: `Exit to Gyroscan Manager menu`

WARNING

If the TCP/IP address of the just deleted DICOM Application is not used by another network application, it has to be removed from the ucx database manually. This is not done automatically, because the DICOM Application doesn't know if this IP address is used elsewhere e.g., it is possible that two DICOM Applications run on the same host.

13. If you have to remove the IP address continue with the next step, otherwise continue with step 19.
14. Select: `Exit to DCL`
15. Enter: **ucx**
16. Enter: `ucx>` **sh host**
17. Enter: `ucx>` **set nohost <hostname>** (in our example : set nohost EASYVISION)
18. Enter: `ucx>` **exit**
19. Enter: **logout** and confirm by clicking on **yes**

8. GENERAL NETWORK INFORMATION

The network is used to transport information (i.e. medical images or patient information) between different modalities in order to share the information across them. You can use an EasyVision CT/MR to view the examinations, which have been made with Gyroscan Intera. Also a DICOM printer can be connected to the network.

The hospital network types can be any of the following:

- Thick ethernet, also called 10Base5.
- Thin ethernet, also called 10Base2.
- TP or twisted pair. Also called 10BaseT or 100BaseT. This cable is available as shielded twisted pair (TP or STP) and unshielded twisted pair (UTP).

NOTE

From R7 onwards, only UTP and/or STP connections are supported by the Gyroscan Intera. Thin ethernet is not supported because no galvanic separation is present between the Gyroscan host and the thin ethernet.

Popular name	Type of network	Medium type	Segment length	Max. total length	Plug type to computer	Data rate
Thicknet	10Base5	Coax (50 Ohm), thick yellow cable with section marks	500 m	2800 m	AUI (D-plug)	10 Mb/s
(U)TP	10BaseT 100BaseT	(un)shielded twisted pair	100 m	500 m	MMJ plug	10 Mb/s 100Mb/s

Table 5: Various network types

NOTE

- *The maximum length of (U)TP cable depends also on the cable quality. Use certified CAT5 cables only, which can be bought locally to meet the specifications mentioned in Table 5.*
- *For thicknet, the termination of one end of the cable must be isolated from ground potential, the termination of the other side must be grounded (earthed) and the transceiver must be isolated from ground.*

Other network interfaces, such as fiber optic repeaters, and media converters such as AUI -> Thin wire are also available on the market. PMS does not provide them.

9. APPENDIX A: HOSPITAL NETWORK LAN CHECKLIST

8 Hospital network connection

Full or Half Duplex . ________
Network speed ________Mb/s

Network Switch:

Node name : ______________
MAC address : ______________
TCP/IP address : ______________

8

7 6 5 4 3 2 1

Host computer:

Node name : ____________
DECNET address : ____________
TCP/IP address : ____________
Network mask · ____________
Broadcast mask : ____________
Internet domain : ____________

Reconstructor:

Node name : ____________
TCP/IP address : ____________
Subnet mask : ____________
Workgroup : ____________

Print server:

Node name : ____________
MAC address : ____________
TCP/IP address : ____________

EasyVision(s):

Node name : ______ / ______
DECNET address : ______ / ______
TCP/IP address : ______ / ______
DICOM Export AE_title : ______ / ______
DICOM Import AE_title : ______ / ______

Dicom HC print unit:

Dicom SPU AE_title : ____________
Dicom SPC AE_title : ____________
TCP/IP address : ____________
Port number : ____________

Dicom RIS unit:

Node name : ____________
Dicom RIS AE_title : ____________
TCP/IP address : ____________
Port number : ____________

10. APPENDIX B: STAND-ALONE LAN DEFAULTS

Network Switch:

Node name : SWITCH
MAC address : AABBCCDDEEFF
TCP/IP address : 192.192.192.212

6 5 4 3 2 1

Host computer: (1)

Node name : NTSCAN
DECNET address : 1.201
TCP/IP address : 192.192.192.201
Network mask : 255.255.255.0
Broadcast mask : 192.192.192.255
Internet domain : MR

Reconstructor: (2)

Node name : RECONSTRUCTOR
TCP/IP address : 192.192.192.210
Subnet mask : 255.255.255.0
Workgroup : WORKGROUP

Print server: (3)

Node name : JETDIRECT
MAC address : BBCCDDEEFFGG
TCP/IP address : 192.192.192.211

EasyVision: (4)

Node name : ____________
DECNET address : ____________
TCP/IP address : ____________
DICOM Export AE_title : ____________
DICOM Import AE_title : ____________

Dicom HC print unit: (5)

Dicom AE_title : ____________
TCP/IP address : ____________
Port number : ____________

Dicom RIS unit: (6)

Node name : ____________
Dicom RIS AE_title : ____________
TCP/IP address : ____________
Port number : ____________